

Tran Quoc Truong

Applied AI Engineer | QA Automation & AI Evaluation

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Summary

Applied AI Engineer with a QA automation and Python backend background, building RAG, document-intelligence, and LLM/agent evaluation systems.

Delivered AI-assisted QA workflows that increased test coverage by approximately 40% and reduced manual analysis effort by up to 70%, with an emphasis on traceable, testable, and reliable AI systems.

Experience

FPT Software, Applied AI Engineer | QA Automation & AI Evaluation

Ho Chi Minh City, Vietnam

Dec 2024 – present

1 year 8 months

- Built AI-assisted QA workflows for document analysis, test-case generation, duplicate-defect detection, and change-impact analysis, increasing test coverage by approximately 40% and reducing manual analysis effort by up to 70% across banking projects.
- Designed Django/DRF and FastAPI services with PostgreSQL, Redis/Celery, and asynchronous workers, separating domain state from long-running AI processing.
- Developed LLM evaluation and RAG validation pipelines with evidence capture, quality gates, and regression reporting to identify inconsistent or low-confidence outputs.
- Built and presented technical demonstrations that contributed to approximately 20 person-months of follow-on delivery work.

FPT Software Academy, Full-Stack Developer Intern

Ho Chi Minh City, Vietnam

Sept 2024 – Dec 2024

4 months

- Built Django-based web applications and containerized services with Docker.
- Standardized deployment configuration and Cloudflare integration, reducing deployment cycle time by approximately 50%.

Selected Projects

Omni-Agent — AI Agent Hub

Independent AI platform for versioned document knowledge workflows and agent evaluation, coordinated through a Django/DRF control plane.

- Designed a project-centric Django/DRF control plane for permissions, document state, job lifecycles, workflow rules, APIs, and artifact lineage.
- Implemented versioned document ingestion, OCR, chunking, and traceable RAG workflows using PostgreSQL and S3-compatible object storage.
- Integrated stateless processing workers while keeping domain state and lifecycle decisions within the main backend.

Core Banking & Payment Hub QA Automation

Client delivery work at FPT Software covering omnichannel core banking and financial transaction integrations across API, database, regression, and automation testing.

- Built and demonstrated an omnichannel automation proof of concept across Web, iOS, and Android, contributing to approximately 20 person-months of follow-on delivery work.
- Validated end-to-end financial transaction flows through API testing, SQL-based database verification, regression testing, and stress testing.
- Authored more than 1,500 lines of reusable SQL validation queries and maintained coverage across all in-scope API endpoints.
- Reduced manual QA effort by approximately 70% through reusable automation and transaction-validation workflows.

ATA — Agent Test Agent

Hackathon MVP for independently validating captured AI-agent runs and producing auditable quality verdicts; its evaluation design later informed Omni-Agent's Agent Testing module.

- Built an end-to-end evaluation platform with a FastAPI control plane, LiteLLM capture gateway, Celery workers, and React/Vite dashboard.
- Designed a multi-stage evaluator using Planner, Judge, Critic, and Meta-Judge roles to assess context use, tool calling, security, and business outcomes.
- Implemented percentile-based quality gates using P10/P35/P50 metrics and blocker dimensions to expose low-tail failures, together with fail-closed isolation of provider credentials.

APIT — Agent Programmatic Intelligence Testing

Graduation project that analyzes API documentation and generates structured API test cases with LLMs.

- Built an AI workflow for API-document understanding, structured test-case generation, and automated output evaluation.
- Evaluated LoRA and QLoRA approaches and fine-tuned Qwen2.5-3B, achieving macro-F1 0.655, a 41.5% relative improvement over a Llama-3.2-3B baseline in the same evaluation setup.
- Implemented FastAPI-based evaluation workflows and LLM-as-a-Judge experiments for comparing generated outputs.

Technical Skills

Applied AI & Evaluation: RAG, Hybrid Retrieval, Agentic Workflows, LLM Evaluation, LLM-as-a-Judge, Evaluation Datasets, Judge/Critic/Meta-Judge Patterns, LoRA/QLoRA

AI Tooling & Observability: LangGraph, LangChain, LiteLLM, Langfuse, Qdrant, Docling, OCR and Document Processing

Backend & API Engineering: Python, Django/DRF, FastAPI, REST APIs, OpenAPI, Asynchronous Workers, Service Layers, Domain Modeling, Job Lifecycle Management

Data, Storage & Infrastructure: PostgreSQL, Redis, Celery, S3/MinIO, Docker, Docker Compose, Nginx, Cloudflare, Presigned URLs

Quality Engineering & Domain: API Validation, SQL Validation, Regression Testing, Test Automation, CI/CD, Git, Core Banking, Payment Hub, E-Commerce

Education

FPT School of Business & Technology, Master of Software Engineering in AI

Ho Chi Minh City, Vietnam
May 2026 – present

FPT University, Bachelor of Artificial Intelligence

Ho Chi Minh City, Vietnam
2021 – 2025

Languages

Vietnamese: Native

English: Working proficiency